Percentage

Percent is a value when checked w.r.t. 100.

like, If you get 80% on a test,
you got 80 out of 100 questions right.

$\Rightarrow$ Three Comparison tools $\rightarrow$

1) Difference
2) Ratio
3) Percentage

examples $\rightarrow$

- A person spends 20% of his salary.

  20% $\Rightarrow$ 20 out of 100

  If salary is 100, expense is 20.

  $20\% = \frac{2\cancel{0}}{10\cancel{0}} = \frac{1}{5}$ $\left\{ 20\% = \frac{1}{5} \right\}$

- If he spends 25% of his salary.

  $25\% = \frac{25}{100} = \frac{1}{4}$ $\left\{ 25\% = \frac{1}{4} \right\}$

  It means, if a whole part is 100,
  we are talking about 25 out of 100
  or we say, one-fourth.

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Ratio → Ratio is a comparision by division

- Boy 1 have 20 chocolates,
  Boy 2 have 30 chocolates, find the ratio.

Boy 1 : Boy 2

$2\cancel{0} : 3\cancel{0}$

$2 : 3$

Or, can be written as,

$$\frac{\text{Boy 1}}{\text{Boy 2}} = \frac{2\cancel{0}}{3\cancel{0}} = \frac{2}{3}$$

{Ratio should be solved in its lowest terms.}

Percentage is also a ratio compared to 100.

Boy : Girl

$4\cancel{0} : 6\cancel{0}$

$\cancel{4} : \cancel{6}$

$2 : 3$

Or $\dfrac{\text{Boy}}{\text{Girl}} = \dfrac{4\cancel{0}}{6\cancel{0}} = \dfrac{4}{6} = \dfrac{2}{3}$

Boys in class = 40

Girls in class = 60

Now,

⇒ Boys contribute to what portion of class ?

whole class = (2+3) = 5

Boys = 2

Ratio = $\dfrac{2}{5}$ → Part of Boys / → whole class

⇒ Girls contribute to what portion of class ?

Ratio = $\dfrac{3}{5}$ (Girls) / (whole class)

⇒ Amit obtained 30 marks out of 75 marks, find his marks percentage.

$75 \longrightarrow 30$

$1 \longrightarrow \frac{30}{75}$

$100 \longrightarrow \left(\frac{30}{75} \times 100\right)$

Here we are taking total marks as 100.

{Percentage is always computed out of 100}

$$\frac{\cancel{30}}{\cancel{75}_{3}} \times \cancel{100}^{4} \Rightarrow \underline{\underline{40\%}}$$

⇒	Chintu	Mintu
Total Salary →	1500	1800
Expenses →	225	270

- Chintu

Percentage $\Rightarrow \frac{225}{15\cancel{00}} \times 1\cancel{00} = 15\%$

- Mintu

Percentage $= \frac{270}{18\cancel{00}} \times 1\cancel{00} = 15\%$

⇒ Exercise →

$15\% = \frac{15}{100} = \frac{3}{20}$

$75\% = \frac{75}{100} = \frac{3}{4}$

$25\% = \frac{25}{100} = \frac{1}{4}$

$35\% = \frac{35}{100} = \frac{7}{20}$

$80\% = \frac{8\cancel{0}}{10\cancel{0}} = \frac{4}{5}$

$45\% = \frac{45}{100} = \frac{9}{20}$

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- $\frac{8}{40} \Rightarrow \frac{8}{4\cancel{0}} \times 10\cancel{0} = 20\%$
- $\frac{5}{25} \Rightarrow \frac{5}{25} \times 100 = 20\%$
- $\frac{8}{50} \Rightarrow \frac{8}{50} \times 100 = 16\%$
- $\frac{9}{45} \Rightarrow \frac{9}{45} \times 100 = 20\%$
- $\frac{16}{80} \Rightarrow \frac{16}{80} \times 100 = 25\%$
- $\frac{4}{10} \Rightarrow \frac{4}{10} \times 100 = 40\%$

- $24\% = (4\%) \times 6$

$= \frac{1}{25} \times 6$

$= \frac{6}{25}$

or

$24\% = 24/100$

$24\% = \frac{\cancel{24}}{\cancel{100}} = \frac{6}{25}$

$24\% = \frac{6}{25}$ ka

25 → 25 is Base value

- $37.5\% = (12.5\%) \times 3$

$= \frac{1}{8} \times 3$

$37.5\% = \frac{3}{8}$ ka

- $62.5\% \Rightarrow 12.5\% \times 5$

$= \frac{1}{8} \times 5$

$62.5\% = \frac{5}{8}$ ka

8 → Base value

- $48\% = 4\% \times 12$

$= \frac{1}{25} \times 12$

$48\% = \frac{12}{25}$

- $22.22\% \Rightarrow (11.11\%) \times 2$

$= \frac{1}{9} \times 2$

$22.22\% = \frac{2}{9}$

- $60\% = 20\% \times 3$

$= \frac{1}{5} \times 3$

$60\% = \frac{3}{5}$

- $36.36\% \Rightarrow 9.09\% \times 4$

$= \frac{1}{11} \times 4$

$36.36\% = \frac{4}{11}$

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- $325\% \Rightarrow 300\% + 25\%$

$= 3 + \frac{1}{4}$

$= \frac{13}{4}$

- $152\% \Rightarrow 100\% + 52\%$

$= 1 + \frac{13}{25}$

$= \frac{38}{25}$

- $502\% \Rightarrow 500\% + 2\%$

$= 5 + \frac{1}{50}$

$= \frac{251}{50}$

- $412.5\% \Rightarrow 400\% + 12.5\%$

$= 4 + \frac{1}{8}$

$= \frac{33}{8}$

- $277.77\% \Rightarrow 200\% + 77.77\%$

$= 2 + \frac{7}{9}$

$= \frac{25}{9}$

- $416.66\% \Rightarrow 400\% + 16.66\%$

$= 4 + \frac{1}{6}$

$= \frac{25}{6}$

- $180\% \Rightarrow 100\% + 80\%$

$= 1 + \frac{4}{5}$

$= \frac{9}{5}$

- $610\% \Rightarrow 600\% + 10\%$

$= 6 + \frac{1}{10}$

$= \frac{61}{10}$

- $260\% \Rightarrow 200\% + 60\%$

$= 2 + \frac{3}{5}$

$= \frac{13}{5}$

- $225\% \Rightarrow 25\% \times 9$

$= \frac{1}{4} \times 9$

$= \frac{9}{4}$

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Percentage Increase / Decrease →

(I) If a number is increased by 20%.

Let Number = 100

Increase = + 20

New = 120

Initial : New

$\cancel{100} : \cancel{120}$

5 : 6

$20\% = \frac{1}{5}$

Initial : New

$5 \xrightarrow{+1} 6$

$\boxed{5 : 6}$

(II) If a Number is increased by 50%.

Let Number = 100

Increase = +50

New = 150

Initial : New

$10\cancel{0} : 15\cancel{0}$

$\cancel{10} : \cancel{15}$

2 : 3

$50\% = \frac{1}{2}$

Initial : New

$2 \xrightarrow{+1} 3$

$\boxed{2 : 3}$

(III) If a Number is increased by 37.5%.

Let Number = 100

Increase = +37.5

New = 137.5

Initial : New

100 : 137.5

8 : 11

$37.5\% = \frac{3}{8}$

ka

Initial : New

$8 \xrightarrow{+3} 11$

$\boxed{8 : 11}$

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(IV) If a Number is increased by 36.36%

Let Number = 100

Increase = +36.36

New = 136.36

$36.36\% = \frac{4}{11}$

Initial : New

100 : 136.36

1 : $\frac{15}{11}$

11 : 15

Initial : New

11 : 15

$\underbrace{11 \to 15}_{+4}$

$\boxed{11 : 15}$

⇒ What is 24% of 2400 ?

$\frac{24}{1\cancel{00}} \times 24\cancel{00} = 24 \times 24$

$= 576$

⇒ What is 16.67% of 1680 ?

$\left\{16.67\% = \frac{1}{6}\right\}$

$\frac{1}{6} \times 1680 = 280$

⇒ 24 is what percent of 80 ?

$\frac{24}{8\cancel{0}} \times 10\cancel{0} = \frac{\cancel{24}^{3}}{\cancel{8}} \times 10$

$= 30\%$

⇒ 29 is what percent of 232?

$\frac{\cancel{29}^{1}}{\cancel{232}_{8}} \times 100 = \frac{100}{8}$

$= 12.5\%$

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